

- 6 (a) concentric circles ...(*at least three lines*)M1
 with increasing separationA1
 correct direction clearB1 [3]
- (b) (i) correct position to left of wireB1 [1]
- (ii) $B = (4\pi \times 10^{-7} \times 1.7) / (2\pi \times 1.9 \times 10^{-2})$ C1
 $= 1.8 \times 10^{-5} \text{ T}$ A1 [2]
- (c) distance $\propto$ currentC1
 current $= (2.8 / 1.9) \times 1.7$
 $= 2.5 \text{ A}$ A1 [2]

[Total: 8]

7 (a) e.g. more (output) power available